

A Design proposal of a Conventional MZI

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Abstract - This proposal presents the design and analysis of a conventional asymmetric Mach-Zehnder interferometer (aMZI) implemented with Electron Beam Lithography (EBL) for fast prototyping in a platform of silicon-on-insulator (SOI) at UBC. The proposed device incorporates a pair of Y-branch splitters or combiners, optical waveguides, and two grating couplers (GC) I/O interface. De-embedding structures for measurement calibration to quantify variability due to manufacturing tolerance is included. A comprehensive analysis of fabrication-induced variability is presented, addressing critical parameters including effective and group refractive indices, waveguide width, thickness, cross-sectional shape, and surface roughness. Mitigation strategies based on variation analysis are finally discussed.

Index Terms - Mach-Zehnder interferometer, silicon-on-insulator, asymmetric interferometer, fabrication variability, design-manufacturing co-optimization, surface roughness, refractive index, group index, Y-branch coupler, Grating coupler

1. INTRODUCTION

The Asymmetric Mach-Zehnder interferometer (aMZI) is a fundamental optical structure widely employed in sensing applications due to its high sensitivity to environmental parameters such as refractive index, temperature, and mechanical stress. By splitting coherent light into two paths and recombining them, MZIs generate interference patterns sensitive to minute phase differences between the paths. This principle has been extensively leveraged for delay line frequency discriminator based phase noise measurement, in a reference oscillator less homodyne receiver.[2]

Silicon-on-insulator (SOI) technology has emerged as a promising platform for photonic integrated circuits due to its high refractive index contrast, compatibility with complementary metal oxide semiconductor (CMOS) processes, and strong optical confinement. However, the performance of SOI based photonic devices is particularly

susceptible to fabrication imperfections including line edge roughness, waveguide dimension variations, and sidewall imperfections, which introduce parasitic losses and phase errors, ultimately degrading device performance and yield. These challenges become increasingly critical in aMZI designs where precise control over optical path differences is essential for predictable operation. For this University of British Columbia (UBC) edX course, Electron-Beam Lithography (EBL) is used for rapid prototyping and high-resolution patterning of two conventional aMZI along with de-embedding structures for measurement calibration to quantify variability due manufacturing tolerance.

2. DEVICE DESIGN AND OPERATING PRINCIPLES

2.1. ASYMMETRIC MZI STRUCTURE

The proposed aMZI device is implemented on an SOI platform with a 220 nm thick silicon device layer and a silicon dioxide (SiO_2) buried oxide layer. As illustrated in Figure 1, the interferometer consists of two arms: a delay arm incorporating a longer loop and a reference arm with a shorter physical path. [1] [3] [4] [5]

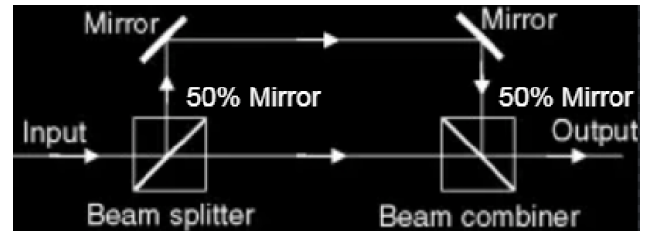


Fig. 1a- An optical bench representation of the aMZI.

The Si integration of this discrete version follows much of the electrical board level to the integrated circuit solution counterpart. For example, the partial mirrors are implemented as a slow radius 90-degree bend, also known as a sweeping bend; a bend commonly used in applications such as aspirating smoke detection systems or electrical conduit [8] and a Bezier bend [7], is generally used for low loss optical waveguide application.

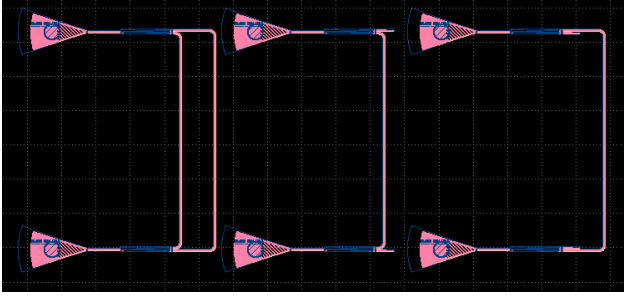


Fig. 1b- The design proposal of the aMZI, de-embedding structures included.

The y-branch 3dB splitter combiner (shown as dark blue instances after the replaces the GC in Fig. 1b) replace the 50% partial mirrors (represented as a box and diagonal) where if an input intensity I_i and electric field E_i the values at the output sides (i.e. I_o and E_o) will correspond to the equal half power split of the input (Eq.1). Similarly, a y-branch combiner will combine modes resulting in an output electric field values (Eq.2) to resolve an output value corresponding to the vector summation of the inputs (Eq.3). Thus, for the lossless case of a plane wave with electric field of and propagation constant (Eq.4)

$$(Eq. 1) I_1 = I_2 = \frac{I_i}{2}, E_1 = E_2 = \frac{E_i}{\sqrt{2}}$$

$$(Eq. 2) E_{o1} = \frac{E_1}{\sqrt{2}}, E_{o2} = \frac{E_2}{\sqrt{2}}$$

$$(Eq. 3) E_o = \frac{(E_{o1} + E_{o2})}{\sqrt{2}}$$

$$(Eq. 4) E = E_o e^{i(\omega t - \beta z)}, \quad \beta = \frac{2\pi n}{\lambda}$$

$$(Eq. 5) E_{o1} = E_1 e^{-i\beta L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_1} (Eq. 6) E_{o2}$$

$$= E_2 e^{-i\beta L_2} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_2}$$

$$(Eq. 7) E_o = \frac{E_i}{\sqrt{2}} \cdot (e^{-i\beta L_1} + e^{-i\beta L_2})$$

$$(Eq. 8) |E_o|^2 = |E_i|^2 \cdot [1 + \cos \beta (L_1 - L_2)]$$

The effective index, n_{eff} (Eq. 10), is dimensionless as the refractive index, but it models light propagation in a waveguide, where the propagation path three boundaries are confined compared to an infinite space. In general terms, n_{eff} is a number that quantifies the phase velocity per unit length in a waveguide, relative to the phase velocity in vacuum. It depends on the wavelength λ and material ambient temperature, and propagation mode and consequently on waveguide geometry [6]. The group index n_g (Eq.11) determines the free-spectral range FSR (Eq.12), an effective refractive index offset versus wavelength and MZI arms' length asymmetry ΔL or pictorially the spacing between adjacent peaks:

$$(Eq. 9) n(\lambda, T) = \frac{c}{v(\lambda, T)}$$

$$(Eq. 10) n_{eff} = \frac{c}{v_p}, n_g = \frac{c}{v_g}$$

$$(Eq. 11) n_g(\lambda) = n_{eff}(\lambda) - \lambda \left[\frac{d}{d\lambda} n_{eff}(\lambda) \right]$$

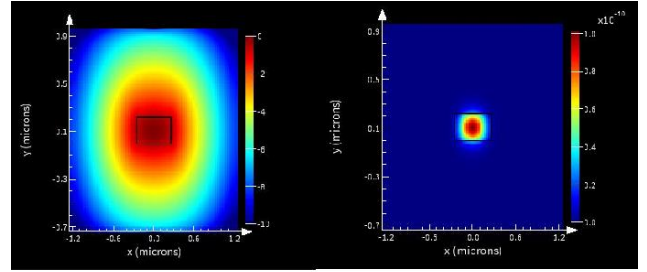
$$(Eq. 12) FSR = \frac{\lambda^2}{\Delta L \cdot n_g(\lambda)}$$

The group velocity, and refractive index are similar in concept to group delay where an envelope of modulated signal experiences differing propagation dependent on the group of frequencies constituting the modulated signal.

Ultimately, a more compact device layout while maintaining the phase sensitivity benefits of a longer interferometer are important steps towards variability mitigation.

2.2. WAVEGUIDE MODEL

The aMZI (Fig 1b) is TE mode optimized therefore the GC is oriented to the left of the layout in accordance with UBC measurement setup requirements to facilitate TM GC access on the right without needing to rotate the setup. The design initially began with mode characterizing the waveguide 500nm by 220nm cross section. (Fig 2)



Field intensity decay at waveguide edges.

Log scale

Energy density; linear scale

Fig. 2a- Field intensity decay at waveguide edges; log scale, Energy density; linear scale

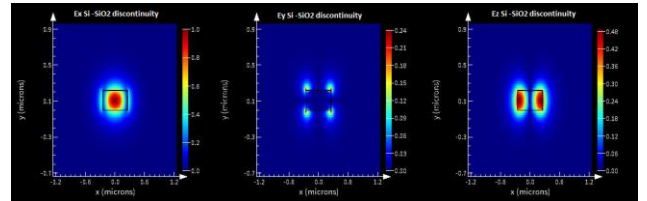


Fig. 2b- E field Si-SiO2 discontinuity

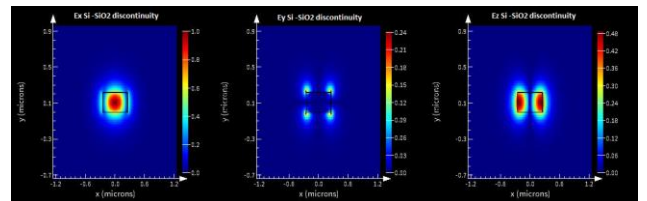


Fig. 2c- H field Si-SiO2 discontinuity

After completing these rigorous mode analyses, an estimate that is reasonably accurate was obtained using the Effective Index Method, a two step process where first, n_{eff} of the slab TE mode for the out of plane cross section is found, second the same procedure is repeated for the in plan cross section. The result from the first step is used as input into the second. The result is the TE $n_{eff} \sim 2.45$ of the 2D waveguide. The inherent assumption is that the fields are sparable/orthogonal, like the separation of variables method for solving differential equations.

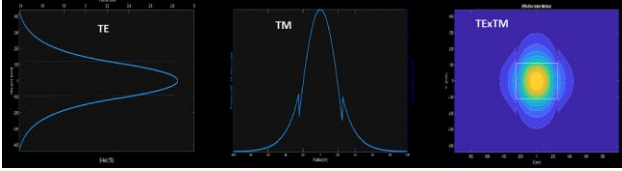


Fig. 2d- Approximate Mode solution: n_{eff} method.

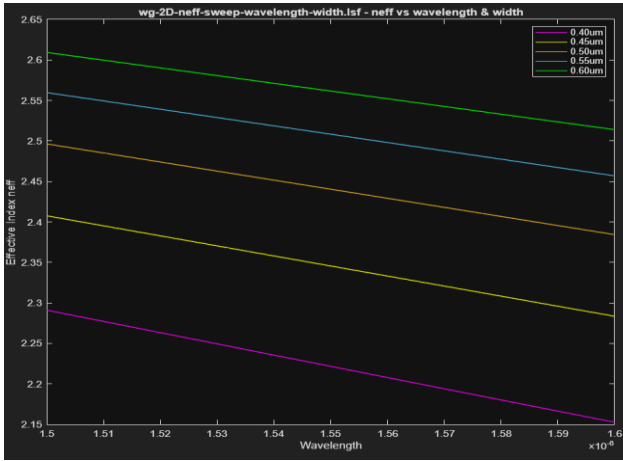


Fig. 2e- Effective Index vs waveguide width.

The data obtained from the wavelength sweep in Lumerical MODE the CML fitting is done with a script in Mode or Matlab code with linear algebra to find a polynomial fit to the simulated n_{eff} data. Fig 2f shows the validity of the fitted polynomial to the original data.



Fig. 2f- CML fitting in Lumerical Mode.

$$(Eq. 9) \quad n_{eff} = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$$

$$(Eq. 9a) \quad n_g = n_{eff}(\lambda) - \lambda \cdot \left(\frac{d}{d\lambda} n_{eff}(\lambda) \right)$$

$$(Eq. 9b) \quad D_{eff} = \lambda^2 \cdot \left(\frac{d^2}{d\lambda^2} n_{eff}(\lambda) \right)$$

In Eq.9, the Taylor expansion centered at the operating wavelength has parameters $n_1 = 2.56162$, $n_2 = -0.95324$, and $n_3 = -0.00378928$. Other optical parameters such as group index (n_g) and dispersion (D_{eff}) can be derived from the effective refractive index (n_{eff}) already modeled and fitted for compact model libraries (CML) component instantiation.

2.3. CIRCUIT MODEL

The aMZI TE network (Fig. 1b) was simulated in Interconnect. All waveguide's (WG) properties such as n_{eff} , n_g , and D were imported into the optical test bench from the Mode CML. The WGs' geometry and material properties being the same except for length difference. An automated flow (Klayout to Interconnect) was used to generate the test bench and initiate the simulation.

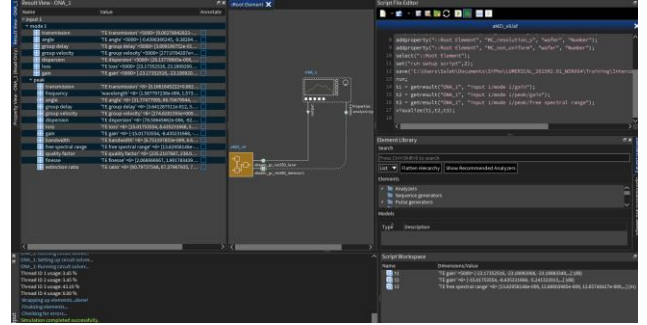


Fig. 3a- EBeam_sosman_v02.gds aMZI_v0 Interconnect test bench.

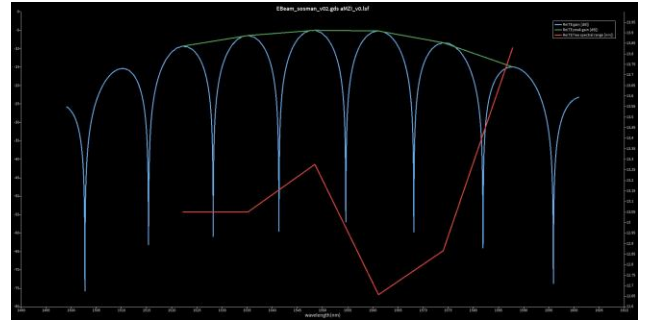


Fig. 3b- aMZI peak/gain [dB] (-5dB) and FSR [nm] (~13nm).

It is worth noting that the Klayout script did not work due to an error loading the netlist element `':design kits::ebeam_dream::ebeam_dream_splitter_1x2_te1550'`, because it is not available in the library. To resolve this issue a simple text edit of the netlist element to point to `':design kits::dream::dream_splitter_1x2_te1550'` was all that was necessary to correct the former error. Another issue was encountered with the paper clip spiral not being supported by a library model which broke the above flow.

To work around the problem, the paper clip path length was simulated with an equivalent length of WG.

3. EBL VARIABILITY ANALYSIS

An integrated photonic circuit (PIC), similar to its electronic counterpart, requires components to match on-chip, wafer-to-wafer, and lot-to-lot. A PIC, particularly, in terms of operational center wavelength, commonly 1550 nm. In implementation, however, it is challenging to achieve the condition that guarantees matching between the two propagation constants β_1 and β_2 of the WGs due to variations in the manufacturing process. The incoming Si material wafer thickness is a variable that must be measured consequently the WG height and milled width. These variations can be gradual and localized on-chip to cause optical yield loss. Critical Dimension (CD) variation of feature profile such as width, edge slope, surface uniformity, and facet roughness is dependent on resist thickness and sensitivity, which can be related to age, exposure, development time, and etching chamber conditions.

There are three main types of variations: systematic that occur all the time such as photo and etch shift, process drift due to metrology or chamber conditions, and random variations. Generally, manufacturing variability can have a multitude of effects depending on the physical layout of the WGs. A corner-analysis flow in the pre-tape-in design review (DR) sign-off offers a worst cases analysis beyond typical. The PIC is typically fabricated on 6 inch, prime-grade wafer from, for example, Soitec, which is sourced with a mean thickness of 219.2 nm and 6-sigma standard of 23.4 nm. Experimental data from UBC EBL, the corner analysis range is in Fig. 4.

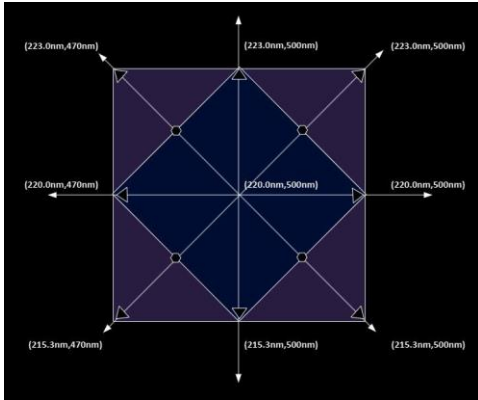


Fig. 4a- WG 9x corners.

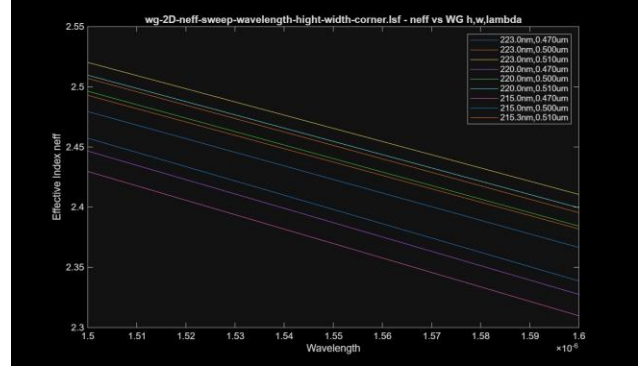


Fig. 4b- WG's n_{eff} versus 9x corners.

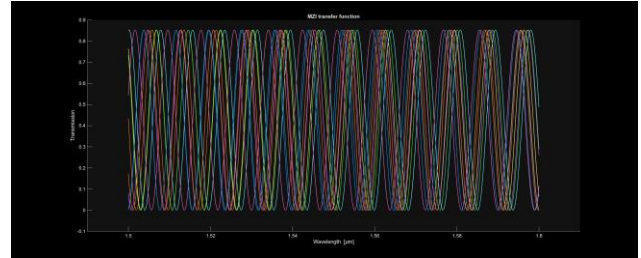


Fig. 4c- aMZI transfer versus 9x corners linear scale.

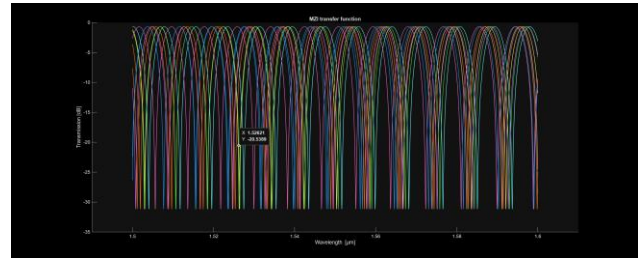


Fig. 4d- aMZI transfer versus 9x corners dB scale.

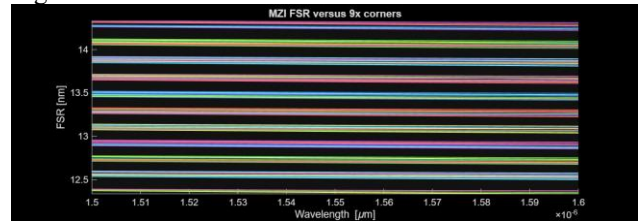


Fig. 4e- aMZI FSR versus 9x corners ~12.37 – 14.33 nm.

The impact of these corner variations may not coincide with all equal probability. Rather, it attempts to set the boundary condition limits and be a substitute for Monte Carol (MC) analysis if manufacturing statistical is not readily available.

4. EBL MEASUREMENT DATA ANALYSIS

A set of images captured from post measurement raw data analysis in Matalab are presented in Fig. 5.

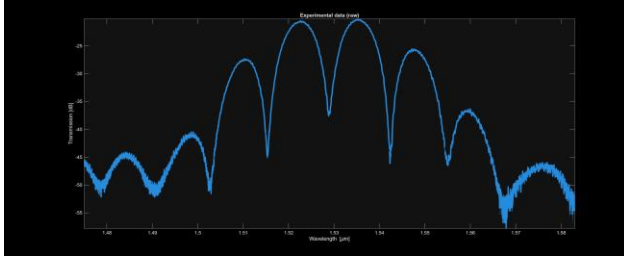


Fig. 5a- aMZI raw measurement data: Transmission [dB].

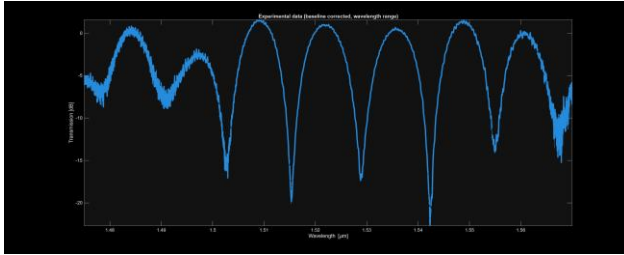


Fig. 5b- Baseline corrected wavelength range.

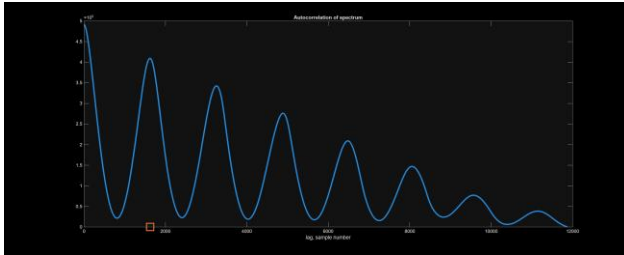


Fig. 5c- Autocorrelation of the spectrum.

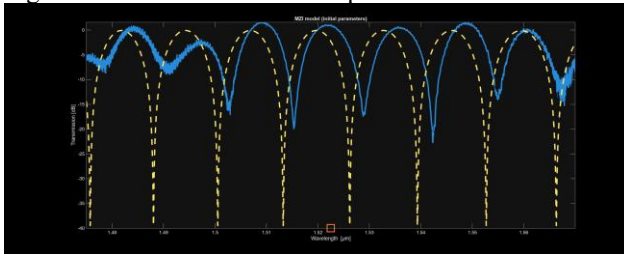


Fig. 5d- Model with initial fitting parameters.

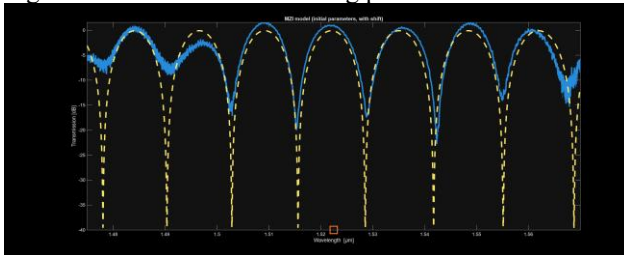


Fig. 5e- Model with initial fitting parameters and shifted.

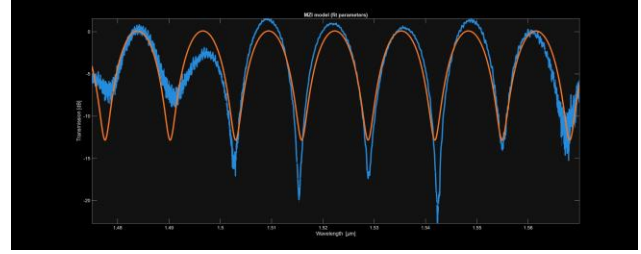


Fig. 5f- Model with final parameters fitting.

The resulting FSR, n_{g0} , and n_{eff} coefficients were:

$$FSR = 12.976 \text{ nm}$$

$$n_{g_av} = 4.1544$$

$$n_{eff}(\lambda) = 2.4072 - 1.1555(\lambda - \lambda_0) + 0.0214(\lambda - \lambda_0)^2$$

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